Sketcbook Upper Bounds

ABSTRACT

CCS CONCEPTS

• **Theory of computation → Bloom filters and hashing**; • **Information systems → Unidimensional range search**.

KEYWORDS

Range Filter, Dynamic Data Structure, Data Growth, Scalability.

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1 STUFF

2 THEORETICAL ANALYSIS AND RESULTS

For simplicity, we will first discuss the analysis for the case in which we have an insertion-only stream, and later discuss why our arguments carry over when we also have deletions. In the following, we will also assume that, for both CS and CMS, the number of arrays we employ is $d = 1$. This does not change the expected error (in the CMS case) or the variance of the error (in the CS case), merely the confidence we have over the estimate.

We will do the analysis assuming a general size function W and then discuss how the analysis proceeds for specific instantiantions of W (linear or sub-linear regime). Note that we perform doubling each time the function W doubles. In the CMS case, W is a function of $N = \sum f(x)$, the ℓ_1 norm of the stream. In the CS case, W is a function of $F = \sum f^2(x)$, the squared ℓ_2 norm of the stream. In either case, we can track the moments *in time* in which the value of the function doubles. Formally, instead of thinking of W as a function of the the ℓ_1 or ℓ_2 norms of the stream, we think of it as a function of time such that, at any point in time t, W_t is the number of counters the sketch uses at time t.

Division of time into epochs. Let T_0 be the first time the sketch is instantiated: this sketch has W_0 counters and is used as long as the length of the stream is at most N_0. We then define T_i for $i \geq 1$ to be the first moment in time when the length of the array doubles, i.e., $W_i = 2 \cdot W_{i-1}$. Note that, in between times T_i and T_{i+1}, the length of the array stays the same (i.e., it is equal to W_i). We call this period of time epoch i and denote by $\mathcal{S}_i$ the *active* sketch during this epoch (i.e., the one that has arrays of length W_i). According to our design, updates to the stream during Epoch i are executed directly on $\mathcal{S}_i$. For the purposes of executing queries fast, we also aggregate all previous sketches into the active sketch.

Formally, let $(x, 1)$ be an update that occured in some epoch $0 \leq j < i$ before the current epoch. In the aggregation step, this update operation is replicated on all the counters in $\mathcal{S}_i$ that x could have hashed into. Namely, if x hashed to an index $\mathcal{J}_x$ in epoch j for one of the arrays, then, during epoch i, all of the 2^{j-1} counters $\{0,1\}^{j-i} \circ \mathcal{J}_x$ ingest the update (x, Δ), where $\circ$ denotes concatenation and $\{0,1\}^{j-i}$ denotes all binary strings of length $j-1$. From the perspective of the fixed counter, what this means is that the counter ingests previous updates from elements that don't technically hash into it. We describe this in more detail in what follows.

Agreggating counters from previous epochs. Let $\mathcal{I}$ be a counter in some epoch i. This counter ingests the following operations:

- updates $(x, 1)$ from epoch i and for which $h(x) = \mathcal{I}$. This event occurs with probability $1/W_i$ since there are W_i counters in the array during that epoch. For ease of analysis, we define $C_{i,\mathcal{I}}(x)$ to be the indicator random variable that is 1 if this event occurs, and 0 otherwise. Then $\Pr\left[C_{i,\mathcal{I}}(x) = 1\right] = 1/W_i$.
- updates $(x, 1)$ from epoch $i-1$ that, during epoch $i-1$, hashed into a counter that matches the rightmost $\log W_i - 1$ bits of $\mathcal{I}$ in its binary representation. This event occurs with probability $2/W_i$. The factor 2 accounts for the mistakes that the aggregation step makes, i.e., processing updates from previous epoch that would not have hashed into $\mathcal{I}$ in the current epoch.
- in general, updates $(x, 1)$ from epoch $j < i$ and for which $h(x)$ matches $\mathcal{I}$ in the rightmost j bits of its binary representation. This event occurs with probability $2^{i-j}/W_i$. Similarly as before, we define $C_{j,\mathcal{I}}(x)$ to be the indicator random variable that is 1 if this event occurs, and 0 otherwise. Then $\Pr\left[C_{j,\mathcal{I}}(x) = 1\right] = 2^{i-j}/W_i$.

While updates from previous epochs have a higher chance of hashing into $\mathcal{I}$ than updates in epoch i, we can nevertheless bound the influence they have on the total error of the estimate returned by CMS, CS and Misra Gries.

Streams with deletions. In order to account for deletions, we note that the state of Sketchbook in a stream with deletions is dominated by the state of Sketchbook on a related insertion-only stream. To begin with, notice that, due to the way we perform contractions, we can pretend that all previous epochs (except the one we're actively in), have only insertions in them (the state of counters in Sketchbook is exactly the same as if we only had insertions previously). We now argue about deletions $(x, -1)$ that only occur during some active epoch i and that have not triggered a contraction. If, during the same epoch i, there is also an insertion $(x, 1)$, then we can act as if this (insertion, deletion) pair did not occur in the stream, i.e., the value of the counter incurs no change during that epoch i. Any extra deletions during epoch i that are not cancelled out by insertions during the same epoch can only decrement the value of the counter, and they do not negatively affect the upper bounds we compute. In

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other words, the value of the counter in epoch i is smaller than its value in a stream in which we remove these extra deletions. Most crucially, by considering this insertions-only stream, we do not miss any errors the counter accumulates from previous epochs. Namely, during epoch i might still incur errors from previous insertions of x from epochs $j < i$ that would not have hashed to the counter during epoch i (i.e., the event $C_{j,\mathcal{I}}(x)$ still holds and we cannot undo it since we do not know when the last insertion of x occured).

2.0.1 Bounds for CMS. In the case of CMS, each counter gathers the sum of frequencies of elements that hash into it and returns this as the estimate. We show the following:

Lemma 2.1. *Let $E_W(N)$ denote the estimate of Sketchbook$_{CMS}$ with size function W and one array of counters, on a stream of ℓ_1 norm N. Then, at the end of epoch i, we have the following:*

$$\mathsf{E}\left[E_W(N_{i+1})\right] \leq \frac{1}{W_i} \cdot \sum_{j=0}^{i} 2^{i-j} \cdot (N_{j+1} - N_j) \,,$$

where N_j for $j \geq 0$ denotes the ℓ_1 norm of the stream at the start of epoch j.

Proof. During epoch i, we have $N_{i+1} - N_i$ fresh insertions (i.e., that have not been cancelled out by deletions). We group all these insertions that occur during a given epoch into one big insertion per element. Formally, let $f_i(x)$ increase in the frequency of x during epoch i. Note that $\sum_x f_j(x) = N_{j+1} - N_j$ for all $j \geq 0$. Note that, in our previous notation, $E_W(N_{i+1}) \leq \mathcal{I}$ and can be expressed as:

$$\mathcal{I} = \sum_{j=0}^{i} \sum_{x} C_{j,\mathcal{I}}(x) \cdot f_j(x) \,.$$

In other words, the counter $\mathcal{I}$ gets the frequency $f_j(x)$ added to it if and only if the event $C_{j,\mathcal{I}}(x)$ occurs. We also have that $\mathsf{E}\left[C_{j,\mathcal{I}}(x)\right] = 2^{i-j}/W_i$. By linearity of expectation, we then get that:

$$\begin{aligned} E[\mathcal{I}] &= \sum_{j=0}^{i} \sum_{x} \frac{2^{i-j}}{W_i} \cdot f_j(x) = \frac{1}{W_i} \cdot \sum_{j=0}^{i} \sum_{x} 2^{i-j} \cdot f_j(x) \\ &= \frac{1}{W_i} \cdot \sum_{j=0}^{i} 2^{i-j} \sum_{x} f_j(x) = \frac{1}{W_i} \cdot \sum_{j=0}^{i} 2^{i-j} \cdot (N_{j+1} - N_j) \,. \end{aligned}$$

The claim follows. □

In other words, the increase in error we get because of aggregating the sketches is dominated by the term $\sum_{j=0}^{i} 2^{i-j} \cdot (N_{j+1} - N_j)$. This is where the specific size function W we employ has an effect, since it translates into bounds on N_j (and specifically their relative sizes). We now consider various special cases and discuss their respective behavior:

Theorem 2.2. *Let $E_W(N)$ denote the estimate of Sketchbook$_{CMS}$ with size function W and one array of counters, on a stream of ℓ_1 norm $N = \sum_y f(y)$. Then:*

$$\mathsf{E}\left[E_W(N)\right] \leq \begin{cases} \log_2(N) \cdot \frac{N}{W(N)}, & \text{if } W(N) = N/\epsilon, \\ \frac{N}{W(N)}, & \text{if } W(N) = N^{1-\alpha}/\epsilon, \text{ for } 0 < \alpha < 1 \end{cases} .$$

Proof. In the case in which $W(N) = N/\epsilon$, the size function doubles whenever the number of elements doubles, i.e., $N_{j+1} = 2 \cdot N_j$ for all $j \geq 0$ and, in general, $N_j = 2^j \cdot N_0$ for all $j \geq 0$. The additional error term then becomes:

$$\begin{aligned} \sum_{j=0}^{i} 2^{i-j} \cdot (N_{j+1} - N_j) &= \sum_{j=0}^{i} 2^{i-j} \cdot (2^{j+1} \cdot N_0 - 2^j \cdot N_0) \\ &= \sum_{j=0}^{i} 2^{i-j} \cdot 2^j \cdot N_0 = (i+1) \cdot 2^i \cdot N_0 \\ &= (i+1) \cdot N_i \,. \end{aligned}$$

Plugging in this bound in the formula from Lemma 2.1 (i.e., dividing by $W(N_i) = W_i$), we get the main claim for $W(N) = N/\epsilon$.

The case in which $W(N) = N^\alpha/\epsilon$, for some $0 < \alpha < 1$. In this case, the size function doubles more rarely, whenever $N_{i+1} = 2^{1/\alpha} \cdot N_i$. Since $0 < \alpha < 1$, note that this implies that N_{i+1} is bigger than $2 \cdot N_i$. This difference is why we avoid the extra dependency on i in the error term. Formally, we have that $N_i = 2^{i/\alpha} \cdot N_0$ and thus:

$$\begin{aligned} \sum_{j=0}^{i} 2^{i-j} \cdot (N_{j+1} - N_j) &= \sum_{j=0}^{i} 2^{i-j} \cdot (2^{(j+1)/\alpha} \cdot N_0 - 2^{j/\alpha} \cdot N_0) \\ &= \sum_{j=0}^{i} 2^{i-j} \cdot 2^{j/\alpha} \cdot N_0 \cdot (2^{1/\alpha} - 1) \\ &= 2^i \cdot N_0 \cdot (2^{1/\alpha} - 1) \cdot \sum_{j=0}^{i} 2^{j/\alpha - j} \,. \end{aligned}$$

For simplicity, we define $2^{1/\alpha - 1} = a$ and observe that:

$$\sum_{j=0}^{i} 2^{j/\alpha - j} = \sum_{j=0}^{i} a^j = (a^{i+1} - 1)/(a-1) \,.$$

Moreover, we have that $2a = 2^{1/\alpha}$ and we can rewrite the error term as:

$$\begin{aligned} \sum_{j=0}^{i} 2^{i-j} \cdot (N_{j+1} - N_j) &= 2^i \cdot N_0 \cdot (2^{1/\alpha} - 1) \cdot \sum_{j=0}^{i} 2^{j/\alpha - j} \\ &= 2^i \cdot N_0 \cdot (2a - 1) \cdot (a^{i+1} - 1)/(a-1) \\ &\leq 2^{i+1} \cdot N_0 \cdot a^{i+2}/(a-1) \,, \end{aligned}$$

where in the last step, we used the bounds $2a - 1 \leq 2a$ and $a^{i+1} - 1 \leq a^{i+1}$ for simplicity. Since $N_{i+1} = (2a)^{i+1} \cdot N_0$, we have that:

$$\sum_{j=0}^{i} 2^{i-j} \cdot (N_{j+1} - N_j) \leq N_{i+1} \cdot a/(a-1) \,.$$

The claim follows by plugging this bound in the expected error from Lemma 2.1.

□

2.1 Bounds for CS

In the case of CS, we obtain a similar claim to that of Lemma 2.1, except as a function of the squared ℓ_2 norm of the stream.

Lemma 2.3. *Let $V_W(F)$ denote the estimate of Sketchbook$_{CS}$ with size function W and one array of counters, on a stream with squared ℓ_2 norm $F = \sum_y f(y)^2$. Then, at the end of epoch i, we have the following:*

$$Var\left[V_W(F_{i+1})\right] \leq \frac{1}{W_i} \cdot \sum_{j=0}^{i} 2^{i-j} \cdot (F_{j+1} - F_j)\,,$$

where F_j denotes the squared ℓ_2 norm of the stream at the start of epoch j.

Proof. We are interested in describing the behavior of the random variable $\hat{f}(z) = \mathcal{I} \cdot \sigma(z)$ for any query z, where $\sigma(z)$ is the associated sign function (for simplicity, we again assume that $d = 1$) and $\mathcal{I}$ is the counter that z hashes into. Using the notation we introduced before, this counter $\mathcal{I}$ can be expressed as:

$$\mathcal{I} = \sum_{j=0}^{i} \sum_{x} C_{j,\mathcal{I}}(x) \cdot f_j(x) \cdot \sigma(x) = \sum_{x} C(x)\,,$$

where $C(x) = \sum_{j=0}^{i} C_{j,\mathcal{I}}(x) \cdot f_j(x) \cdot \sigma(x)$ denotes the contribution to counter $\mathcal{I}$, throughout the epochs, of updates to x.

We begin by computing the expected value of the estimator $\hat{f}(z)$ as in the classical CS analysis that shows that in expectation, $\mathcal{I} \cdot \sigma(z)$ is an unbiased estimator, i.e. $E[\mathcal{I} \cdot \sigma(z)] = f(z)$. In the following, we let $f_j(x)$ denote the frequency of x when restricted to epoch j. By linearity of expectation

$$\mathsf{E}[\hat{f}(z)] = \sum_{x \neq z} \mathsf{E}\left[C(x) \cdot \sigma(z)\right] + \mathsf{E}\left[C(z)\sigma(z)\right] = \sum_{j=0}^{i} f_j(z) = f(z)\,,$$

where the last step uses the fact that $\mathsf{E}\left[\sigma(x) \cdot \sigma(z)\right] = 0$ for any $x \neq z$ and that $\Pr\left[C_{j,\mathcal{I}}(z)\right] = 1$ since $h(z) = \mathcal{I}$. We will now bound the variance of this estimator, and in particular, we will be interested in how much bigger it becomes when compared to $\sqrt{F_i/W_i}$. We get that:

$$\begin{aligned}
\mathsf{Var}\left[\hat{f}(z)\right] &= \mathsf{E}\left[\left(\sum_x C(x)\sigma(z)\right)^2\right] - (f(z))^2 \\
&= \sum_x \mathsf{E}\left[(C(x))^2\right] + \sum_x \sum_{y \neq x} \mathsf{E}\left[C(x)C(y)\right] - (f(z))^2 \\
&= \sum_x \mathsf{E}\left[(C(x))^2\right] - (f(z))^2\,,
\end{aligned}$$

where we use the fact that $\mathsf{E}\left[\sigma(x) \cdot \sigma(y)\right] = 0$ for any $x \neq y$ to establish that $\mathsf{E}\left[C(x)C(y)\right] = 0$. We now bound $\mathsf{E}\left[(C(x))^2\right]$. Recall that we defined $C(x) = \sum_{j=0}^{i} C_{j,\mathcal{I}}(x) \cdot f_j(x) \cdot \sigma(x)$. Then:

$$\begin{aligned}
\mathsf{E}\left[(C(x))^2\right] &= \mathsf{E}\left[\left(\sum_{j=0}^{i} C_{j,\mathcal{I}}(x) \cdot f_j(x)\right)^2\right] \\
&= \sum_{j=0}^{i} \mathsf{E}\left[\left(C_{j,\mathcal{I}}(x) \cdot f_j(x)\right)^2\right] + \\
&+ 2 \cdot \sum_{j=0}^{i-1} \sum_{k=j+1}^{i} \mathsf{E}\left[C_{j,\mathcal{I}}(x) \cdot f_j(x) \cdot C_{k,\mathcal{I}}(x) \cdot f_k(x)\right] \\
&= \sum_{j=0}^{i} f_j(x)^2 \cdot \mathsf{E}\left[C_{j,\mathcal{I}}(x)\right] + \\
&+ 2 \cdot \sum_{j=0}^{i-1} \sum_{k=j+1}^{i} f_j(x) \cdot f_k(x) \cdot \mathsf{E}\left[C_{j,\mathcal{I}}(x) \cdot C_{k,\mathcal{I}}(x)\right]\,.
\end{aligned}$$

For bounding the $\mathsf{E}\left[C_{j,\mathcal{I}}(x)\right]$, we can use the fact that $\mathsf{E}\left[C_{j,\mathcal{I}}(x)\right] \leq 2^{i-j}/W_i$. For $\mathsf{E}\left[C_{j,\mathcal{I}}(x) \cdot C_{k,\mathcal{I}}(x)\right]$, we note that the product of the two indicator random variables is 1 only when both of them are 1. Having said that, the two random variables are not independent: if $C_{k,\mathcal{I}}(x) = 1$, then $C_{j,\mathcal{I}}(x) = 1$ for any $k \geq j$. Namely, if during epoch k, x hashed into a counter that was a prefix of $\mathcal{I}$, then the same would have happened during epoch $j \leq k$. We get that:

$$\mathsf{E}\left[C_{j,\mathcal{I}}(x) \cdot C_{k,\mathcal{I}}(x)\right] = \mathsf{E}\left[C_{k,\mathcal{I}}(x)\right]\,.$$

The whole sums becomes:

$$\begin{aligned}
\sum_{j=0}^{i-1} \sum_{k=j+1}^{i} f_j(x) \cdot f_k(x) \cdot \mathsf{E}\left[C_{k,\mathcal{I}}(x)\right] &= \sum_{k=1}^{i} \sum_{j=0}^{k-1} f_j(x) \cdot f_k(x) \cdot \mathsf{E}\left[C_{k,\mathcal{I}}(x)\right] \\
&= \sum_{k=1}^{i} f_k(x) \cdot \mathsf{E}\left[C_{k,\mathcal{I}}(x)\right] \cdot \sum_{j=0}^{k-1} f_j(x) \\
&= \sum_{k=1}^{i} f_k(x) \cdot \mathsf{E}\left[C_{k,\mathcal{I}}(x)\right] \cdot f_{\leq k-1}(x)
\end{aligned}$$

where, for simplicity, we denoted $f_{\leq i}(x) = \sum_{j=0}^{i} f_i(x)$ the sum of frequencies accumulated by x through epochs 0 to i and set $f_{\leq -1}(x) = 0$. Then:

$$\begin{aligned}
\mathsf{E}\left[(C(x))^2\right] &= \sum_{j=0}^{i} \left(f_j(x)^2 + 2f_j(x) \cdot f_{\leq j-1}(x)\right) \cdot \mathsf{E}\left[C_{j,\mathcal{I}}(x)\right] \\
&= \sum_{j=0}^{i} \left(f_{\leq j}(x)^2 - f_{\leq j-1}(x)^2\right) \cdot \mathsf{E}\left[C_{j,\mathcal{I}}(x)\right]\,.
\end{aligned}$$

While the term $f_{\leq j}(x)^2 - f_{\leq j-1}(x)^2$ might seem curious, we note that $\sum_x f_{\leq j}(x)^2 - f_{\leq j-1}(x)^2 = F_j - F_{j-1}$ denotes the increase in the squared ℓ_2 norm of the stream vector as we move from epoch $j-1$ to epoch j. We thus get that:

$$\begin{aligned}\mathsf{Var}\left[\hat{f}(z)\right] &= \sum_x \mathsf{E}\left[(C(x))^2\right] - (f(z))^2 \\ &= \sum_x \sum_{j=0}^{i} \left(f_{\le j}(x)^2 - f_{\le j-1}(x)^2\right) \cdot \mathsf{E}\left[C_{j,\mathcal{I}}(x)\right] - (f(z))^2 \\ &\le \sum_{j=0}^{i} \left(F_j - F_{j-1}\right) \cdot \frac{2^{i-j}}{W_i} \ .\end{aligned}$$

The claim follows. □

At this point, it becomes perhaps more natural to understand why we defined the size function to depend on the squared ℓ_2 norm of the stream vector. We now discuss different bounds we get on the variance depending on the size function.

Theorem 2.4. *Let $V_W(F)$ denote the estimate of Sketchbook$_{CS}$ with size function W and one array of counters, on a stream with squared ℓ_2 norm $F = \sum_y f(y)^2$. Then:*

$$Var\left[V_W(F)\right] \le \begin{cases} \log_2(F) \cdot \frac{F}{W(F)}, & \text{if } W(F) = F/\epsilon, \\ \frac{1-2^{-1/\alpha}}{1-2^{1-1/\alpha}} \frac{F}{W(F)}, & \text{if } W(F) = F^{1-\alpha}/\epsilon, \text{ for } 0 < \alpha < 1 \end{cases} .$$

We note that the calculations from the above are highly similar to those of Sketchbook$_{\text{CMS}}$ from Theorem 2.2, with the difference that we now see a dependency on F rather than N. We include them for completeness:

Proof. In the case in which $W(F) = F/\epsilon$, we start with the original sketch $\mathcal{S}_0$ that allows streams for which their squared ℓ_2 norm is at most some F_0. The size function doubles whenever the squared ℓ_2 norm doubles. In this case $F_{j+1} = 2 \cdot F_j$ for all $j \ge 0$ and, in general, $F_j = 2^j \cdot F_0$ for all $j \ge 0$. In this case, we get that:

$$\begin{aligned}\mathsf{Var}\left[V_W\right] &\le \sum_{j=0}^{i} \left(F_j - F_{j-1}\right) \cdot \frac{2^{i-j}}{W_i} \\ &\le \frac{1}{W_i} \cdot \left(F_0 \cdot 2^i + \sum_{j=1}^{i} \left(2^j - 2^{j-1}\right) \cdot F_0 \cdot 2^{i-j}\right) \\ &\le \frac{1}{W_i} \cdot \left(F_0 \cdot 2^i + \sum_{j=1}^{i} 2^{j-1} \cdot F_0 \cdot 2^{i-j}\right) \\ &\le \frac{1}{W_i} \cdot \left(F_0 \cdot 2^i + \sum_{j=1}^{i} F_0 \cdot 2^{i-1}\right) \\ &\le \frac{1}{W_i} \cdot \left(F_i + (i-1) \cdot F_i/2\right) = \frac{F_i}{W_i} \cdot \frac{i+1}{2} \ .\end{aligned}$$

Setting $W(F) = F^\alpha/\epsilon$ in CS, for some $0 < \alpha < 1$. In this case, the size function doubles more rarely, whenever $F_{i+1} = 2^{1/\alpha} \cdot F_i$. Since $0 < \alpha < 1$, note that this implies that F_{i+1} is bigger than $2F_i$. This difference is why we avoid the extra dependency on i in the error term. Plugging this into the statement of Lemma 2.3, we have that $F_i = 2^{i/(1-\alpha)} \cdot F_0$ and thus:

$$\begin{aligned}\sum_{j=0}^{i} 2^{i-j} \cdot \left(F_{j+1} - F_j\right) &= \sum_{j=0}^{i} 2^{i-j} \cdot \left(2^{(j+1)/(1-\alpha)} \cdot T_0 - 2^{j/(1-\alpha)} \cdot T_0\right) \\ &= \sum_{j=0}^{i} 2^{i-j} \cdot 2^{j/(1-\alpha)} \cdot T_0 \cdot \left(2^{1/(1-\alpha)} - 1\right) \\ &= 2^i \cdot T_0 \cdot \left(2^{1/(1-\alpha)} - 1\right) \cdot \sum_{j=0}^{i} 2^{j/(1-\alpha)-j} \ .\end{aligned}$$

□

2.2 Bounds for Misra Gries

In the case of the MG sketch, we consider a slightly more general division of time into epochs. We start with a sketch that uses $W(N_0)$ counters until the stream length reaches N_0. At this point in time, the sketch sets the number of counters to $W(N_1)$. We leave the relationship between $W(N_0)$ and $W(N_1)$ undefined for now. The sketch uses $W(N_1)$ counters until the stream length reaches N_1, and we call this period of time epoch 1. Once N_2 is reached, the sketch switches to $W(N_2)$ counters (starting epoch 2) and so on. We then prove the following:

Lemma 2.5. *Let $E(N)$ denote the error of Sketchbook$_{MG}$ with size function W on a stream of total key count N. Then, at the end of epoch i, we have the following:*

$$E(N_i) \le \sum_{j=0}^{i} \frac{N_j - N_{j-1}}{W(N_j) + 1} \ ,$$

where N_j for $j \ge 0$ denotes the total key length of the stream at the end of epoch j.

Proof. We begin by first defining some notation. Let C_i denote the total count of frequencies stored in the sketch at the end of epoch i (with $C_{-1} = 0$). Note that $C_i \ge 0$ always, since we only delete from the sketch when all the counters are nonzero. Let d_i also denote the number of deletions we make to the sketch during epoch i. By design, note the following relationship for $i \ge 1$:

$$C_i = C_{i-1} + N_i - N_{i-1} - d_i \cdot (W(N_i) + 1) \ .$$

For simplicity, consider the case in which $i = 1$. At the beginning of epoch i, the frequencies stored in the sketch add up to C_{i-1}. During the epoch, we see $N_1 - N_0$ fresh insertions, each of which either increases the frequencies stored in the sketch by 1, or deletes $W(N_i)$ frequencies from the sketch. The number of elements in $N_1 - N_0$ that each issue these $W(N_i)$ deletes is exactly d_i. In other words, there are $N_1 - N_0 - d_i$ elements that each increases the sketch counts by 1, and d_i elements that each decreases the sketch count by $W(N_i)$. The total changes are therefore $N_1 - N_0 - d_i - d_i \cdot W(N_i)$, which is how we get the first expression. From this, we get that:

$$d_i = \frac{N_i - N_{i-1} + C_{i-1} - C_i}{W(N_i) + 1} \ .$$

Now, for any specific key x, the maximum error that we can make in its frequency is if we delete one from its frequency every time we delete something from the sketch (i.e., this worst-case is achieved when the sketch stores all occurences of x from the beginning and then keeps deleting one from its count every time it

seems an unstored element). Formally, if we are at the end of epoch i, then the error is bounded as:

$$E(N_i) \le f(x) - \hat{f}(x) \le \sum_{j=0}^{i} d_j \ .$$

We now plug in our expression for d_j and use $W_j = W(N_j)$ for ease of notation:

$$\begin{aligned}
\sum_{j=0}^{i} d_j &= \sum_{j=0}^{i} \frac{N_j - N_{j-1} + C_{j-1} - C_j}{W_j + 1} \\
&= \sum_{j=0}^{i} \frac{N_j - N_{j-1}}{W_j + 1} + \sum_{j=0}^{i} \frac{C_{j-1} - C_j}{W_j + 1} \\
&= \sum_{j=0}^{i} \frac{N_j - N_{j-1}}{W_j + 1} + \sum_{j=0}^{i-1} C_j \cdot \left(-\frac{1}{W_j + 1} + \frac{1}{W_{j+1} + 1} \right) - \frac{C_i}{W_j + 1} \\
&= \sum_{j=0}^{i} \frac{N_j - N_{j-1}}{W_j + 1} - \sum_{j=0}^{i-1} C_j \cdot \frac{W_{j+1} - W_j}{(W_j + 1)(W_{j+1} + 1)} - \frac{C_i}{W_i + 1} \\
&\le \sum_{j=0}^{i} \frac{N_j - N_{j-1}}{W_j + 1} \ ,
\end{aligned}$$

where the last inequality holds because $W_{j_1} > W_j$, so we are ignoring negative factors in the sum. □

Before we prove our main lemma for this section, we require the following technical claim:

Claim 2.6. *For any constant $\alpha < 1$, the following holds:*

$$\sum_{j=0}^{i} \frac{2^{j/\alpha} - 2^{(j-1)/\alpha}}{2^j} = O(1) \cdot \frac{2^{i/\alpha}}{2^i} \ .$$

Proof. Note that:

$$\begin{aligned}
\frac{2^{j/\alpha} - 2^{(j-1)/\alpha}}{2^j} &= (1 - 2^{-1/\alpha}) \cdot \frac{2^{j/\alpha}}{2^j} \\
&= (1 - 2^{-1/\alpha}) \cdot \frac{2^{i/\alpha}}{2^i} \cdot \frac{2^{(j-i)/\alpha}}{2^{j-i}} \\
&= (1 - 2^{-1/\alpha}) \cdot \frac{1}{2^{(i-j)\cdot(1/\alpha - 1)}} \cdot \frac{2^{i/\alpha}}{2^i} \ .
\end{aligned}$$

Summing everything up, we get that:

$$\begin{aligned}
\sum_{j=0}^{i} \frac{2^{j/\alpha} - 2^{(j-1)/\alpha}}{2^j} &\le (1 - 2^{-1/\alpha}) \cdot \sum_{j=0}^{i} \frac{1}{2^{(i-j)\cdot(1/\alpha - 1)}} \cdot \frac{2^{i/\alpha}}{2^i} \\
&= O(1) \cdot \frac{2^{i/\alpha}}{2^i} \ ,
\end{aligned}$$

since $1/\alpha > 1$ and the sum $\sum_{j=0}^{i} \frac{1}{2^{(i-j)\cdot(1/\alpha-1)}}$ converges to a constant. □

We are now ready to prove the main bounds on Sketchbook$_{\text{MG}}$

Theorem 2.7. *Letting $E(N)$ denote the error of Sketchbook$_{MG}$ with size function $W(\cdot)$ on a stream with a total key count N, it holds that*

$$E(N) \le \begin{cases} O(1) \cdot \epsilon N^{1-\alpha} & \text{if } W(N) = N^\alpha/\epsilon, \text{ for } \alpha \in [0, 1), \\ \log_2 N \cdot \epsilon & \text{if } W(N) = N/\epsilon. \end{cases}$$

Compared to a fixed-size MG allocated upfront with an array of $W(N)$ counters with knowledge of N, the expected error terms above are higher by at most a constant and a logarithmic factor.

Proof. In the case in which $W(N) = N/\epsilon$, we use the fact that in Sketchbook$_{\text{MG}}$ we increase the counters one by one (rather than double them). In other words, we consider divisions into epochs for which $W(N_{i+1}) = W(N_i) + 1$ and hence $N_{i+1} = N_i + \epsilon$, and so, plugging this into Lemma 2.5, we would have that, at the end of some epoch i:

$$\begin{aligned}
E(N_i) &\le \sum_{j=0}^{i} \frac{N_j - N_{j-1}}{W(N_j) + 1} = \frac{N_0}{W(N_0)} + \sum_{j=1}^{i} \frac{\epsilon}{W(N_0) + j} \\
&= \epsilon + \epsilon \cdot \sum_{j=1}^{i} \frac{1}{W(N_0) + j} \le \epsilon + \epsilon \cdot \int_{W(N_0)+1}^{W(N_0)+i} \frac{1}{x}\, dx \\
&\le \epsilon + \epsilon \cdot \ln\left(\frac{W(N_0) + i}{W(N_0) + 1} \right) \le \epsilon \cdot (1 + \ln i) \le \epsilon \cdot \log_2 N \ ,
\end{aligned}$$

since i is now defined as the number of times we expand by one counter until we reach a stream length of N, and so $i \le N/\epsilon + 1$.

The case in which $w(N) = N^\alpha/\epsilon$. In this case, it is easier to think of divisions into time in which we just double the number of counters. Namely, we define the epochs such that $W(N_{i+1}) = 2 \cdot W(N_i)$ and hence $N_{i+1} = 2^{1/\alpha} \cdot N_i$ (notice that the statement of Lemma 2.5 does not depend on how specifically the epochs are defined). Then we have that the sum satisfies:

$$\begin{aligned}
\sum_{j=0}^{i} \frac{N_j - N_{j-1}}{W(N_j) + 1} &= \frac{2^{j/\alpha} \cdot N_0 - 2^{(j-1)/\alpha} \cdot N_0}{2^j \cdot W(N_0)} \\
&= \sum_{j=0}^{i} \frac{2^{j/\alpha} - 2^{(j-1)/\alpha}}{2^j} \cdot \frac{N_0}{W(N_0)} \ .
\end{aligned}$$

We now apply Claim 2.6 and get that:

$$\sum_{j=0}^{i} \frac{N_j - N_{j-1}}{W(N_j) + 1} \le O(1) \cdot \frac{2^{i/\alpha}}{2^i} \cdot \frac{N_0}{W(N_0)} = O(1) \cdot \frac{N_i}{W(N_i)} \ .$$

The main claim follows. □

2.3 Misra Gries with residuals

Lemma 2.8. *Let $E(N)$ denote the error of Sketchbook$_{MG}$ with size function W on a stream of total key count N. Then, at the end of epoch i, we have the following:*

$$E(N_i) \le \sum_{j=0}^{i} \frac{N_j^{res(k)} - N_{j-1}^{res(k)}}{W(N_j) + 1 - k} \ ,$$

where $N_j^{res(k)}$ for $j \ge 0$ denotes the total key length of the stream at the end of epoch j minus the frequencies of the top k elements in the stream by the end of epoch i.

Proof. We start similarly as in the proof of Lemma 2.5 and get that, if d_i is the number of deletions to the sketch that occur during some epoch $i \ge 0$, then:

$$C_i = C_{i-1} + N_i - N_{i-1} - d_i \cdot (W(N_i) + 1) \ ,$$

where C_i denotes the total count of frequencies stored in the sketch at the end of epoch i (with $C_{-1} = 0$). We further can claim that:

$$C_i - C_{i-1} \geq N_i^{(k)} - N_{i-1}^{(k)} - dk \ ,$$

where $N_i^{(k)}$ denotes the frequencies of the top k elements at the end of epoch i (the top k with respect to the whole stream). In other words, the change in the sketch frequencies is at least the change in the sketch frequencies restricted to the top k elements: the total frequency restriced to these k elements increases by $N_i^{(k)} - N_{i-1}^{(k)}$ and each of these k elements gets its frequency reduced at most d_i times. We then you get that:

$$d_i \leq \frac{N_i - N_{i-1} - N_i^{(k)} + N_{i-1}^{(k)}}{W(N_i) + 1 - k} = \frac{N_i^{\text{res}(k)} - N_{i-1}^{\text{res}(k)}}{W(N_i) + 1 - k} \ .$$

Adding up, we get the claim.

□

We now proceed to show a similar bound as before. Namely:

Theorem 2.9. *Letting $E(N)$ denote the error of Sketchbook$_{MG}$ with size function $W(\cdot)$ on a stream with a total key count N, it holds that*

$$E(N) \leq \begin{cases} O(1) \cdot \epsilon \left(N^{res(k)}\right)^{1-\alpha} & \text{if } W(N) = N^\alpha/\epsilon, \text{ for } \alpha \in [0,1), \\ \epsilon \cdot \log_2 N^{res(k)} & \text{if } W(N) = N/\epsilon, \end{cases}$$

where $N^{res(k)}$ denotes the total key length of the stream minus the frequencies of the top k elements in the stream.

Proof. In the case in which $W(N) = N/\epsilon$, we have that $W(N_{i+1}) = W(N_i) + 1$ and $N_{i+1} = N_i + \epsilon$. Fix an epoch i. Define i_1 as the largest index such that

$$\sum_{j=1}^{i_1} \left(N_j^{\text{res}(k)} - N_{j-1}^{\text{res}(k)}\right) \leq \epsilon \ .$$

In other words, we group consecutive terms of the form $N_j^{\text{res}(k)} - N_{j-1}^{\text{res}(k)}$ until we are about to reach ϵ. Note that

$$\sum_{j=1}^{i_1+1} \left(N_j^{\text{res}(k)} - N_{j-1}^{\text{res}(k)}\right) \geq \epsilon \ ,$$

since the extra term $N_{i_1+1}^{\text{res}(k)} - N_{i_1}^{\text{res}(k)} \leq \epsilon$. Similarly, we define $i_2 > i_1$ to be the largest index such that

$$\sum_{j=i_1+1}^{i_2} \left(N_j^{\text{res}(k)} - N_{j-1}^{\text{res}(k)}\right) \leq \epsilon \ .$$

Similarly as before, we stop right before we gather a sum of ϵ, and thus:

$$\sum_{j=i_1+1}^{i_2+1} \left(N_j^{\text{res}(k)} - N_{j-1}^{\text{res}(k)}\right) \geq \epsilon \ ,$$

We continue as such for ℓ steps until we have that:

$$\sum_{j=i_{\ell+1}}^{i} \left(N_j^{\text{res}(k)} - N_{j-1}^{\text{res}(k)}\right) \leq \epsilon \ .$$

We now define $i_0 = 0$ and get ℓ can be upper bounded as such:

$$\begin{aligned} N_i^{\text{res}(k)} &\geq \sum_{j=1}^{i_\ell} \left(N_j^{\text{res}(k)} - N_{j-1}^{\text{res}(k)}\right) \\ &\geq \frac{1}{2} \cdot \sum_{s=0}^{\ell=1} \sum_{j=i_s+1}^{i_{s+1}+1} \left(N_j^{\text{res}(k)} - N_{j-1}^{\text{res}(k)}\right) \\ &\geq \frac{1}{2} \cdot \sum_{s=0}^{\ell=1} \epsilon = \epsilon \cdot \ell/2 \ . \end{aligned}$$

In other words, $\ell \leq 2N_i^{\text{res}(k)}/\epsilon$. Now we also notice that for every $0 \leq s \leq \ell$ with $i_{\ell+1} = i$, it holds that

$$\begin{aligned} \sum_{j=i_s+1}^{i_{s+1}} \frac{N_j^{\text{res}(k)} - N_{j-1}^{\text{res}(k)}}{W(N_j) + 1 - k} &\leq \sum_{j=i_s+1}^{i_{s+1}} \frac{N_j^{\text{res}(k)} - N_{j-1}^{\text{res}(k)}}{W(N_0) + s + 1 - k} \\ &= \frac{\epsilon}{W(N_0) + s + 1 - k} \ . \end{aligned}$$

This is because $W(N_j)+1-k = W(N_0)+j+1-k \geq W(N_0)+s+1-k$, since $j \geq i_s + 1 \geq s$. We thus get that:

$$\begin{aligned} \sum_{j=0}^{i} \frac{N_j^{\text{res}(k)} - N_{j-1}^{\text{res}(k)}}{W(N_j) + 1 - k} &\leq \frac{N_0^{\text{res}(k)}}{W(N_0) + 1 - k} + \sum_{s=0}^{\ell} \frac{\epsilon}{W(N_0) + s + 1 - k} \\ &\leq \epsilon + \epsilon \cdot \sum_{s=0}^{\ell} \frac{1}{W(N_0) + s + 1 - k} \\ &\leq \epsilon + \epsilon \cdot \int_{W(N_0)+1-k}^{W(N_0)+\ell+1-k} \frac{1}{x} dx \\ &\leq \epsilon + \epsilon \cdot \ln \ell \leq \epsilon \cdot \log_2 N_i^{\text{res}(k)} \ . \end{aligned}$$

The case in which $w(N) = N^\alpha/\epsilon$. In this case, it is easier to think of divisions into time in which we just double the number of counters. Namely, we define the epochs such that $W(N_{i+1}) = 2 \cdot W(N_i)$ and hence $N_{i+1} = 2^{1/\alpha} \cdot N_i$. We also define:

$$E_i' = \sum_{\ell=0}^{f(i)} \frac{2^{\ell/\alpha} \cdot N_0^{\text{res}(k)} - 2^{(\ell-1)/\alpha} \cdot N_0^{\text{res}(k)}}{2^\ell \cdot W\left(N_0^{\text{res}(k)}\right) - k + 1} \ ,$$

where $f(i)$ is defined as the smallest integer such that $2^{f(i)/\alpha} \cdot N_0^{\text{res}(k)} \geq N_i^{\text{res}(k)}$. We thus also have that

$$2^{f(i)/\alpha} \cdot N_0^{\text{res}(k)} \leq 2^{1/\alpha} \cdot N_i^{\text{res}(k)} \ .$$

Note that $f(0) = 0$ and that, in spirit, E' represents an upper bound on the error we would incur if we were to double our counters as a function of $N^{\text{res}(k)}$ rather than N. We then claim that for all $i \geq 0$:

Claim 2.10.

$$\sum_{j=0}^{i} \frac{N_j^{res(k)} - N_{j-1}^{res(k)}}{W(N_j) + 1 - k} \leq 4 \cdot E_i'$$

Before we prove Claim 2.10, we would first like to discuss how it implies our bounds. Namely, we first note that each denominator can be substituted by $2^\ell \cdot W\left(N_0^{\text{res}(k)}\right)$. Then:

$$E'_i \le \sum_{\ell=0}^{f(i)} \frac{2^{\ell/\alpha} \cdot N_0^{\mathrm{res}(k)} - 2^{(\ell-1)/\alpha} \cdot N_0^{\mathrm{res}(k)}}{2^\ell \cdot W\left(N_0^{\mathrm{res}(k)}\right)}$$
$$= \sum_{\ell=0}^{f(i)} \frac{2^{\ell/\alpha} - 2^{(\ell-1)/\alpha}}{2^\ell} \cdot \frac{N_0^{\mathrm{res}(k)}}{W\left(N_0^{\mathrm{res}(k)}\right)}$$
$$= O(1) \cdot \frac{2^{f(i)/\alpha} \cdot N_0^{\mathrm{res}(k)}}{2^{f(i)} \cdot W\left(N_0^{\mathrm{res}(k)}\right)} ,$$

where the last equality holds by Claim 2.6. We now note that, by definition, the numerator $2^{f(i)/\alpha} \cdot N_0^{\mathrm{res}(k)} \le 2^{1/\alpha} \cdot N_i^{\mathrm{res}(k)}$ and the denominator satisfies:

$$2^{f(i)} \cdot W\left(N_0^{\mathrm{res}(k)}\right) = \left(2^{f(i)/\alpha} \cdot N_0^{\mathrm{res}(k)}\right)^\alpha / \epsilon$$
$$\ge \left(N_i^{\mathrm{res}(k)}\right)^\alpha / \epsilon = W\left(N_i^{\mathrm{res}(k)}\right) .$$

Therefore:

$$E'_i = O(1) \cdot \frac{N_i^{\mathrm{res}(k)}}{W\left(N_i^{\mathrm{res}(k)}\right)} .$$

In conjunction with Lemma 2.8, we get the main claim for $W(N) = N/\epsilon$. We now prove the remaining claim.

Proof of Claim 2.10. For $i = 0$, note that $f(0) = 0$ and so:

$$\frac{N_0^{\mathrm{res}(k)}}{W(N_0) + 1 - k} = E'_0$$

For every $i \ge 1$, we will prove that:

$$\frac{N_i^{\mathrm{res}(k)} - N_{i-1}^{\mathrm{res}(k)}}{W(N_i) + 1 - k} \le 2 \cdot \sum_{\ell=f(i-1)}^{f(i)} \frac{2^{\ell/\alpha} \cdot N_0^{\mathrm{res}(k)} - 2^{(\ell-1)/\alpha} \cdot N_0^{\mathrm{res}(k)}}{2^\ell \cdot W\left(N_0^{\mathrm{res}(k)}\right) - k + 1} .$$

We first upper bound the denominators on the righthand side as:

$$2^{f(i)} \cdot W\left(N_0^{\mathrm{res}(k)}\right) - k + 1 \le 2 \cdot (W(N_i) + 1 - k) .$$

This is because:

$$2^{f(i)} \cdot W\left(N_0^{\mathrm{res}(k)}\right) = 2^{f(i)} \cdot \left(N_0^{\mathrm{res}(k)}\right)^\alpha / \epsilon = \left(2^{f(i)/\alpha} \cdot N_0^{\mathrm{res}(k)}\right)^\alpha / \epsilon$$
$$\le \left(2^{1/\alpha} \cdot N_i^{\mathrm{res}(k)}\right)^\alpha / \epsilon$$
$$\le \left(2^{1/\alpha} \cdot N_i\right)^\alpha / \epsilon = 2 \cdot W(N_i) .$$

For the numerators, we notice that:

$$\sum_{\ell=f(i-1)}^{f(i)} 2^{\ell/\alpha} - 2^{(\ell-1)/\alpha} = \left(2^{f(i)/\alpha} - 2^{(f(i-1)-1)/\alpha}\right) .$$

Now we get that, by definition of $f(i)$, we have that $2^{f(i)/\alpha} \cdot N_0^{\mathrm{res}(k)} \le 2^{1/\alpha} \cdot N_i^{\mathrm{res}(k)}$ and $2^{(f(i-1)-1)/\alpha} \cdot N_0^{\mathrm{res}(k)} \ge N_{i-1}^{\mathrm{res}(k)}$, and so the numerators satisfy:

$$\sum_{\ell=f(i-1)}^{f(i)} \left(2^{\ell/\alpha} \cdot N_0^{\mathrm{res}(k)} - 2^{(\ell-1)/\alpha} \cdot N_0^{\mathrm{res}(k)}\right) \ge N_i^{\mathrm{res}(k)} - N_{i-1}^{\mathrm{res}(k)} .$$

Putting everything together, we have that:

$$\frac{N_i^{\mathrm{res}(k)} - N_{i-1}^{\mathrm{res}(k)}}{W(N_i) + 1 - k} \le \sum_{\ell=f(i-1)}^{f(i)} \frac{2^{\ell/\alpha} \cdot N_0^{\mathrm{res}(k)} - 2^{(\ell-1)/\alpha} \cdot N_0^{\mathrm{res}(k)}}{W(N_i) + 1 - k}$$
$$\le 2 \cdot \sum_{\ell=f(i-1)}^{f(i)} \frac{2^{\ell/\alpha} \cdot N_0^{\mathrm{res}(k)} - 2^{(\ell-1)/\alpha} \cdot N_0^{\mathrm{res}(k)}}{2^\ell \cdot W\left(N_0^{\mathrm{res}(k)}\right) - k + 1} ,$$

which is what we wanted to prove for every specific epoch i. Summing things up over all epochs, and noting that some terms get doublecounted, we get the main claim.

□

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